

CONSERVATION



THE PRINCIPLE AIM of conservation is to ensure the survival of life in all its forms and variety, and to make certain that natural resources are not used beyond their capacity for renewal but continue for the benefit of future generations. Conservation requires an understanding of ecology – the interrelationships of the different plants and animals (including our own species) with each other, and with their environment. Concern for the health of the environment is steadily increasing, as can be seen from the growth of conservation organizations on almost every continent.

Endangered species

As human numbers increase, more land is needed to grow food, so forests are cut down and habitats destroyed. Without its habitat, wildlife cannot survive. Human pressure and hunting is causing many species, such as the white rhino, to become rare or endangered, some to the point of extinction. It is too late for animals such as the Tasmanian wolf, but others, such as the grey whale, have been saved from extinction.



Rhinos are hunted for their horns.

White rhino

Conservation

Conservation means wise use of resources, thus recognizing that to use natural resources is perfectly acceptable, as long as they are not exploited beyond their capacity for renewal. Conservation is concerned with the survival of life in all its forms, and with maintaining organic life at the optimum rate of productivity.

Preservation

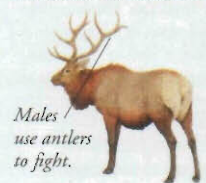
Preservation differs from conservation in that preservation means strict protection as an end in itself, without regard for the consequences. The first area of land to be preserved was Yellowstone National Park in the USA, where animals such as grizzly bears and wapitis thrive in this undisturbed environment. Over-protection can lead to a build-up of the animal numbers, causing habitat destruction and the starvation and decline of the animals. This can be avoided by good management and by culling excess numbers of animals.

Yellowstone National Park, USA



Grizzly bears stand 3 m (10 ft) high.

Grizzly bears are the largest and most powerful of the animals living in Yellowstone National Park.



Males use antlers to fight.

Wapitis live in the forests of Yellowstone. They are larger than the red deer of Eurasia, but their behaviour is similar.



When displaying, male lifts tail into a fan.

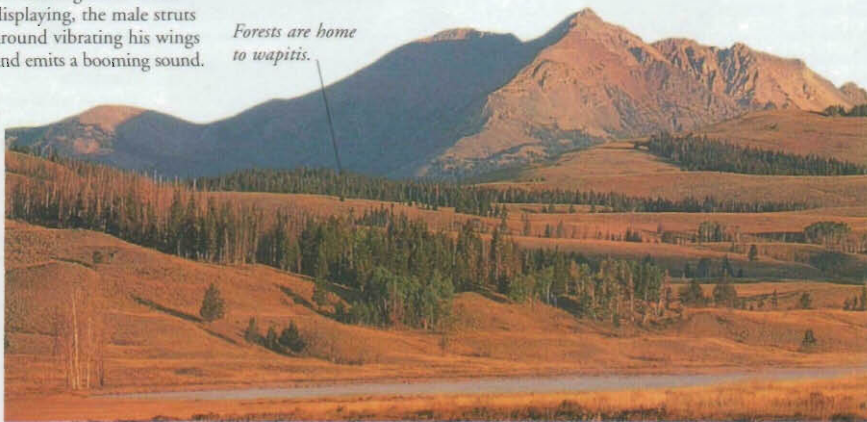
Sage grouse are the most spectacular of the North American grouse. When displaying, the male struts around vibrating his wings and emits a booming sound.



Wingspan may reach 9 cm (3.5 in).

Phoebe butterflies appear in the Rockies in mid-summer.

Forests are home to wapitis.



Environmental organizations

Hundreds of organizations are concerned with safeguarding the Earth's resources. The International Union for Conservation of Nature and Natural Resources (IUCN), and the Worldwide Fund for Nature (WWF), carry out conservation projects worldwide.



Park rangers, WWF, Tsavo National Park

Methods of conservation

The establishment of national parks and wildlife sanctuaries is a very effective method of conserving natural areas and their wildlife. Other methods include education, breeding programmes, using renewable energy such as solar, wind, and wave power, and legislation. Some developing countries have agreed to safeguard areas of natural habitat in return for a reduction in their foreign debt.

Legislation

The Convention on International Trade in Endangered Species (CITES) controls trade in rare species, such as tigers; other groups control fishing and pollution.



Tiger bones are used to make oriental medicines.

Siberian tiger remains

Education

The importance of educating young people about conservation and the effect it has on their lives cannot be overstated. The need for education is as vital in the west as in the developing world. For conservation to be effective, it must have the support of the local people.



Game ranger teaching children

Breeding programmes

The best chance of survival for some animals close to extinction lies in breeding them in captivity for eventual return to their natural habitat. In one of the earliest breeding programmes, the few remaining Arabian oryx were captured, bred in the USA, and later successfully re-introduced to Oman.



Arabian oryx

Pale fur helps conceal oryx in the desert.

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